

Avi Vyas

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PROFESSIONAL EXPERIENCE

Software Engineer (Backend Engineer)

07/2025 – Present | Noida, India

To The New (TTN) 🔗

- Developed and maintained scalable backend microservices using **Java** and **Spring Boot** for a platform serving **18M+ users**.
- Optimized **Elasticsearch-based search and autocomplete APIs**, reducing response time from **1s+ to ~80ms** while achieving **1,500+ TPS** with **~50% of production infrastructure**, significantly improving search performance, throughput, and infrastructure efficiency.
- **Developing an AI-powered Root Cause Analysis (RCA) microservice** that detects **KPI threshold breaches** and generates **evidence-based root-cause reports** from application code, logs, and runtime data, designed for adoption across multiple teams.
- Regularly contribute to **production performance optimisations and incident RCA**, investigating alerts, bottlenecks, and service-level issues across microservices and implementing targeted improvements to **reliability, latency, and system efficiency**.
- **Built a shared backend service** for automated subscriber upgrades, enabling eligible subscribers to upgrade at the same subscription price across **multiple consumer services**, eliminating duplicate implementations.
- **Designed and developed configurable promotions** using **business-rule-driven configuration**, supporting recurring and multiple promotion types while reducing dependency on code changes for new campaigns.

SKILLS

Languages — Java (Primary), Python

Backend & Frameworks — Spring Boot, Spring AI, Spring Security, Spring Data JPA, Hibernate, FastAPI

Databases & Messaging — MySQL, PostgreSQL, Redis, Elasticsearch, RabbitMQ

Cloud & DevOps — AWS, Docker, Git, GitLab, Maven, Gradle

AI & Engineering — MCP, Spec Kit, JUnit, Mockito, Multithreading, OOPS, SOLID, Data Structures, REST APIs, Microservices, Design Patterns

PROJECTS

E-Commerce Backend System

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Java, Spring Boot, MySQL, Spring Security, JWT

- **Built a production-style e-commerce backend with 50+ REST APIs**, implementing **JWT authentication and role-based access control** for Admin, Seller, and Customer workflows.
- **Designed metadata-driven product and category management**, supporting flexible category hierarchies, product variations, and custom attributes.
- **Implemented secure authentication flows** with access/refresh tokens, BCrypt password hashing, account activation, and password reset.

Doctor Appointment System

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Python, FastAPI, MySQL, SQLAlchemy, JWT

- **Built a FastAPI-based appointment management system with JWT authentication and role-based access control** for doctors and patients.
- **Designed availability and appointment scheduling with 30-minute slot generation, conflict detection, and appointment lifecycle management**.
- **Implemented asynchronous database access using SQLAlchemy and MySQL**, with a modular API and service architecture.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering

2021 – 2025

BML Munjal University

CGPA: 8.14 / 10